We claim:

1. A pipeline traversing robot (100) comprising:

a base (15) having three arms extending radially outward from a central hub, forming a three-armed symmetric structure configured at 120 degrees from one another to wrap around the exterior surface of a pipeline such that wheels (10) mounted at the terminal ends of each arm maintain simultaneous contact with the pipe surface from multiple radial directions;

a plurality of screws (11) positioned at the terminal ends of each arm of the base (15);

a plurality of neodymium magnets (3) fixed on the underside of the base (15) at a predetermined clearance from the pipe surface, configured to generate magnetic adhesion force to maintain stable contact with the exterior surface of metallic gas pipes during both horizontal and vertical traversal;

a motor driver (8) comprising of a plurality of motors (7), each electrically connected to the motor driver (8) and mechanically coupled to a respective wheel (10) being configured to provide directional movement of the robot along the pipeline;

a castor wheel (9) configured for 360-degree rotational maneuverability of the robot on the pipeline surface;

a gas sensor (2), a temperature and humidity sensor (6), an ultrasonic sensor (5) and torch (14) mounted on the base (15)

an imaging device (4) mounted on the base (15) and configured to capture image data of the surrounding pipeline environment for remote visual inspection;

a monitoring and control center (12);

a data store (13); and

a processing unit (1) electrically connected to the gas sensor (2), temperature and humidity sensor (6), ultrasonic sensor (5), imaging device (4), motor driver (8), and torch (14), and configured to receive sensor data, process the data stored in the data

store (13), and transmit leakage status and alerts to the monitoring and control center (12).

2. A pipeline traversing robot (100) as claimed in claim 1, wherein the plurality of neodymium magnets (3) are fixed on the underside of each arm of the base (15) proximal to the terminal end thereof, such that the magnetic attraction force generated between the magnets (3) and the exterior metallic surface of the gas pipe maintains inward radial contact pressure of the wheels (10) against the pipe surface , configured for adhesion of robot to the pipe along with continuous traction of the wheels (10) during traversal.

3. A pipeline traversing robot (100) as claimed in claim 1, wherein the plurality of screws (11) positioned at the terminal ends of the three arms forms a wrapping grip configuration around the exterior surface of pipes having diameters in the range of 150 mm to 200mm.

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